

Multiple Choice Questions (80 points: 20 questions at 4 points each)

1. What is the primary role of the bias term (w_0) in a linear model?
 - A. Control model complexity
 - B. Shift the decision boundary parallel to itself
 - C. Scale the feature values
 - D. Rotate the decision boundary

2. How does the perceptron model relate to logistic regression?
 - A. Both use a generalized linear model framework, but perceptron has a hard boundary
 - B. Both models are types of neural networks
 - C. Perceptron is an unsupervised version of logistic regression
 - D. Both use a soft decision boundary to classify data

3. The perceptron learning algorithm is guaranteed to converge only under what condition?
 - A. For linearly separable training data
 - B. When using stochastic gradient descent
 - C. When feature scaling is applied
 - D. For datasets with large sample size

4. Which best describes bagging in the context of random forests?
 - A. Combining trees by training them sequentially
 - B. Sequentially improving weak learners
 - C. Training multiple models on subsets sampled with replacement
 - D. Reducing the number of trees to avoid overfitting

5. A decision tree selects the best split at each node using:
 - A. A greedy algorithm
 - B. Random selection
 - C. Gradient descent
 - D. Backpropagation

6. Suppose the computed linear combination (z) of weights and features equals zero. In logistic regression, this indicates that the feature vector x is located:
- A. Directly on the decision boundary
 - B. Definitely in class 0
 - C. Definitely in class 1
 - D. Far from the decision boundary
7. Which of the following is true about "sampling with replacement" in random forests?
- A. Data points selected once cannot be selected again
 - B. Every data point appears exactly once in every subset
 - C. The same data point may appear multiple times in a subset
 - D. Sampling without replacement is mandatory in random forests
8. Which of the following best distinguishes regression trees from model trees?
- A. Regression trees use averages at leaves; model trees fit linear models at leaves
 - B. Model trees use entropy; regression trees use Gini
 - C. Model trees are limited to binary splits; regression trees are not
 - D. Regression trees use numeric features, model trees use categorical
9. Which activation function is typically used by the simple perceptron model?
- A. ReLU (Rectified Linear Unit)
 - B. Heaviside step function
 - C. Sigmoid
 - D. Hyperbolic tangent
10. What is a "weak learner" in the context of ensemble methods?
- A. A fully pruned decision tree
 - B. A model with perfect classification accuracy
 - C. A model that only uses numeric features
 - D. A model slightly better than random guessing

11. In random forests used for regression, the final prediction for a new data point is usually:
- A. The mode of individual tree predictions
 - B. The mean of individual tree predictions
 - C. The weighted majority of individual tree predictions
 - D. The median of individual tree predictions
12. In logistic regression, if you move through feature space exactly opposite the direction of the weight vector, the probability of class 1:
- A. Oscillates between 0 and 1
 - B. Decreases toward 0
 - C. Remains unchanged
 - D. Always increases rapidly toward 1
13. Entropy, in the context of decision trees, is used to:
- A. Normalize feature values
 - B. Measure uncertainty in classification labels
 - C. Calculate distances between feature vectors
 - D. Prune redundant branches in the tree
14. In a decision tree, a split is:
- A. A leaf that contains only one class
 - B. The division of a node based on a feature value
 - C. A measure of model performance on validation data
 - D. A summary of prediction error
15. Which statement about multilayer perceptrons (MLPs) is true?
- A. They cannot learn non-linear decision boundaries
 - B. They are not considered linear models
 - C. They use only the identity activation function
 - D. They are generalized linear models

- * 16. What is the role of an activation function in a perceptron?
- A. To prevent overfitting during training
 - B. To transform the input data into a higher dimension
 - C. To determine the output class based on the weighted sum of inputs
 - D. To add bias to the decision process
17. What is the main goal of a split in a decision tree?
- A. To ensure each node contains all class labels
 - B. To reduce class purity
 - C. To increase the number of features
 - D. To partition data into more homogeneous groups
18. In random forests, each decision tree typically uses:
- A. A random subset of features at each split
 - B. Only numeric features at each split
 - C. All available features for every split
 - D. Only categorical features at each split
19. In logistic regression, traveling parallel to the decision boundary causes the predicted class probabilities to:
- A. Gradually increase
 - B. Stay the same
 - C. Switch abruptly between classes
 - D. Gradually decrease
20. Which of the following is NOT true for logistic regression?
- A. Its decision boundary is linear
 - B. It produces probabilities of class membership
 - C. It directly estimates numeric target values
 - D. It uses a linear combination of features